# Hints: Differentiable functions – 2

Class work

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

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## Problem 1

**Hint 1.** The hypothesis says that $h = f - g'$ takes values of opposite signs at 0 and 1. But $h$ need not be continuous, since $g'$ need not be.

**Hint 2.** Write $h$ as a derivative: $h = F'$ with $F(x) = \int_0^x f(t)\,dt - g(x)$. By Darboux' theorem $h$ has the intermediate value property, so it vanishes somewhere in $(0,1)$.

## Problem 2

**Hint 1.** The equation $f' - f = \alpha$ suggests the multiplier $e^{-x}$: consider $g(x) = e^{-x}\big(f(x) + \alpha\big)$, so that $g' = e^{-x}(f' - f - \alpha)$.

**Hint 2.** $g(0) = \alpha$ and $g(1) = (1+\alpha)/e$. If they are equal, Rolle gives $\xi$; this happens exactly for $\alpha = \frac{1}{e-1}$.

**Hint 3.** For any other $\alpha$, choose $g$ linear, $g(x) = \alpha + \big(\frac{1+\alpha}{e} - \alpha\big)x$, and put $f(x) = e^x g(x) - \alpha$. Then $f \in V$ and $f' - f - \alpha = e^x g'$ never vanishes. Answer: $\alpha = \frac{1}{e-1}$ only.

## Problem 3

**Hint 1.** Put $g(x) = f(x)\cos x$. It vanishes at $-\frac{\pi}{2}$, 0 and $\frac{\pi}{2}$, so Rolle gives $a \in (-\frac{\pi}{2}, 0)$ and $b \in (0, \frac{\pi}{2})$ with $g'(a) = g'(b) = 0$.

**Hint 2.** Now find a multiplier $\varphi$ such that $(\varphi g')'$ contains no $f'$ term. Since $g' = f'\cos x - f\sin x$, the choice $\varphi = 1/\cos^2 x$ works.

**Hint 3.** Check that $\Big(\dfrac{g'(x)}{\cos^2 x}\Big)' = \dfrac{1}{\cos x}\Big(f''(x) - f(x)\big(1 + 2\tan^2 x\big)\Big)$, and apply Rolle on $[a, b]$.

## Problem 4

**Hint 1.** The factor $\frac{4}{\pi} = \frac{1}{\arctan 1 - \arctan 0}$ and $1 + \xi^2 = 1/\arctan'(\xi)$ suggest Cauchy's mean value theorem.

**Hint 2.** Apply it to $f$ and $g = \arctan$ on $[0,1]$: $\dfrac{f(1) - f(0)}{\pi/4} = \dfrac{f'(\xi)}{1/(1+\xi^2)}$.

## Problem 5

**Hint 1.** Take logarithms (everything is positive): the claim becomes $\dfrac{f'(\xi)}{f(\xi)} = \ln f(1) - \ln f(0)$.

**Hint 2.** This is the Lagrange mean value theorem for $\ln f$ on $[0,1]$.